

For $n = 2, m = 77^2 = 5929$

For $n = 3, m = 77^3$
= Last digit will be 3

For $n = 4, m = 77^4$
= Last digit will be 1.

Now, the same pattern of unit digit will be obtained for other values of n .

Thus, we observe that unit digit of m is 1 after each 4 values of n .

Hence, the required probability = $\frac{1}{4}$

110. Option (d) is correct.

$$\begin{aligned} n(S) &= {}^{15}C_3 \\ \therefore 2 \times 3 &= 6 \\ 2 \times 4 &= 8 \\ 2 \times 5 &= 10 \\ 2 \times 6 &= 12 \\ 2 \times 7 &= 14 \\ 3 \times 4 &= 12 \end{aligned}$$

So, we have 6 cases.

$$\Rightarrow n(E) = 6$$

So, the required probability

$$\begin{aligned} P(E) &= \frac{n(E)}{n(S)} = \frac{6}{{}^{15}C_3} \\ &= \frac{6}{\frac{15 \times 14 \times 13}{6}} \\ &= \frac{36}{15 \times 14 \times 13} \end{aligned}$$

$$\Rightarrow P(E) = \frac{6}{455}$$

Solution for Q.no. 111 and 112:

Given that, $P(A \cup B) \geq 0.75$ and $0.125 \leq P(A \cap B) \leq 0.375$

111. Option (d) is correct.

$$\begin{aligned} P(A \cup B) &\geq 0.75 \\ \Rightarrow P(A) + P(B) - P(A \cap B) &\geq 0.75 \\ \Rightarrow P(A) + P(B) &\geq 0.75 + P(A \cap B) \\ \Rightarrow P(A) + P(B) &\geq 0.75 + 0.125 \\ &\quad \{\because P(A \cap B) \geq 0.125\} \\ \Rightarrow P(A) + P(B) &\geq 0.875 \end{aligned}$$

Minimum value of

$$P(A) + P(B) = 0.875$$

112. Option (b) is correct.

$$\begin{aligned} \text{Since, } P(A \cap B) &\leq 0.375 \\ \Rightarrow P(A) + P(B) - P(A \cap B) &\leq 0.375 \\ \Rightarrow P(A) + P(B) &\leq 0.375 + P(A \cap B) \\ \Rightarrow P(A) + P(B) &\leq 0.375 + 0.75 \end{aligned}$$

$$\{\because P(A \cup B) \geq 0.75\}$$

$$\Rightarrow P(A) + P(B) \leq 1.125$$

$$\Rightarrow \text{Maximum value of } P(A) + P(B) \text{ is } 1.125.$$

Solution for Q.no. 113 and 114:

$$\begin{aligned} P(A) &= 0.6, \quad P(B) = 0.4, \quad P(C) = 0.5, \\ P(A \cup B) &= 0.8, \quad P(A \cap C) = 0.3, \quad P(A \cap B \cap C) = 0.2 \\ \text{and } P(A \cup B \cup C) &\geq 0.85 \end{aligned}$$

$$\begin{aligned} P(A \cap B) &= P(A) + P(B) - P(A \cup B) \\ &= 0.6 + 0.4 - 0.8 = 0.2 \end{aligned}$$

$$\begin{aligned} \text{Now, } P(A \cup B \cup C) &= P(A) + P(B) + P(C) - P(A \cup B) \\ &\quad - P(A \cap C) - P(B \cap C) + P(A \cap B \cap C) \\ &= 1.5 - 0.2 - 0.3 - P(B \cap C) + 0.2 \end{aligned}$$

$$\Rightarrow P(A \cup B \cup C) = 1.2 - P(B \cap C)$$

$$\text{Since, } 0.85 \leq P(A \cup B \cup C) \leq 1$$

$$\Rightarrow 0.85 \leq 1.2 - P(B \cap C) \leq 1$$

$$\Rightarrow -0.35 \leq -P(B \cap C) \leq -0.2$$

$$\Rightarrow 0.2 \leq P(B \cap C) \leq 0.35$$

113. Option (b) is correct.

$$\text{Minimum value of } P(B \cap C) = 0.2$$

114. Option (c) is correct.

$$\text{Maximum value of } P(B \cap C) = 0.35$$

Solution for Q.no. 115 and 116:

p = Probability of getting at least one tail.

$$= 1 - \left(\frac{1}{2}\right)^n$$

q = Probability of getting at least two tails.

$$= 1 - \left(\frac{1}{2}\right)^n - \frac{n}{2^n}$$

115. Option (b) is correct.

$$p - q = \frac{5}{32}$$

$$\Rightarrow 1 - \left(\frac{1}{2}\right)^n - \left[1 - \left(\frac{1}{2}\right)^n - \frac{n}{2^n}\right] = \frac{5}{32}$$

$$\Rightarrow \frac{n}{2^n} = \frac{5}{32}$$

$$\text{or } \frac{n}{2^n} = \frac{5}{2^5}$$

Comparing both sides, we get, $n = 5$

116. Option (a) is correct.

$$p + q = 2 - 2\left(\frac{1}{2}\right)^n - \frac{n}{2^n}$$

$$= 2 - 2 \times \frac{1}{2^5} - \frac{5}{2^5}$$

$$= \frac{57}{32}$$